

Numerical supplement

Saved discrepancy evidence

Plymouth conditions are [remove years, adjust]; Brest conditions 2 and 3 mean reduced adjusted and reduced naive. All seven rows failed the original test; the final column shows the separate reference error, not a replacement result. Values are displayed to nine decimals; original JSON determines status.

Table S1. Primary discrepancies and separate reference-optimizer values.

Dataset	Condition	Quantity	Primary	Published	Absolute error	Limit	Reference	Reference error
Plymouth	[False, True]	par0	128.776231113	128.770000000	0.006231113	0.005100000	128.771317149	0.001317149
Plymouth	[False, True]	return100	215.478293518	216.000000000	0.521706482	0.510000000	215.510400474	0.489599526
Plymouth	[True, True]	par1	17.657292743	17.650000000	0.007292743	0.005100000	17.653982487	0.003982487
Plymouth	[True, True]	se0	4.605521290	4.600000000	0.005521290	0.005100000	4.604730745	0.004730745
Plymouth	[True, False]	se0	4.544831541	4.550000000	0.005168459	0.005100000	4.545716410	0.004283590
Brest	2	scale σ	11.924808852	11.930000000	0.005191148	0.005100000	11.925930995	0.004069005
Brest	3	scale σ	12.004301678	12.010000000	0.005698322	0.005100000	12.007526508	0.002473492

Reference evidence uses the unmodified R 4.4.2 nmmin body with local platform shims and the independent density, not a full R package execution. Fixed quartile/zero-shape starts, maxit=500 and relative tolerance $\sqrt{\text{machine epsilon}}$ were used. All reference optimizer status codes were zero. These receipts establish the numerical explanation, not the exact original author environment. They were not rerun for this editorial pass.

Sources: [Plymouth primary](#), [Plymouth reference](#), [Brest primary](#), [Brest reference](#).

Computational settings

Python 3.12, NumPy 2.3.5, SciPy 1.18.1. Point fitting uses Nelder-Mead with maxfev=5000 per start, xatol=1e-8, fatol=1e-9; three quartile-based starts request shape 0, -0.2 and +0.2, capped toward zero where required for initial support. Converged starts must agree in objective within 1e-5. Profile nuisance fits use maxfev=5000, xatol=1e-7, fatol=1e-8, with a zero-shape fallback for an infeasible profile start. Bracketing allows 20 expansions; Brent allows 50 iterations with xtol=1e-5; endpoint likelihood residual must be at most 0.001.

Standard errors use central finite-difference observed information, step sizes 1e-5 times $\max(1, |\mu|)$, $\max(1, \sigma)$, and 1 for shape; the half-step calculation must change each standard error by at most 0.001 and information must be positive definite. These are local checks, not certificates of global optimality. Numerical details are inspectable in [replay.py](#), [audit_endpoints.py](#) and the original application scripts/receipts.

Distinct counts and historical validation

The 120 scalar comparisons comprise parameters, standard errors, return levels and interval endpoints across eight configurations. Original outcomes remain 113 pass and seven fail. The 48 printed endpoints are a subset of these 120. The 96 subsequent solver checks instead use two independent nuisance starts per endpoint; they are not 96 additional published quantities or an acceptance score.

The earlier packaged local chain completed both aggregations/fits and Plymouth's 48 checks, then stopped after 42 Brest checks at the former 10 CPU second audit limit. Only the remaining six checks were completed in a separately capped continuation. At that historical checkpoint the default cap had changed to 15 CPU seconds without a full-chain rerun. Original stopped and continuation receipts are preserved. A separate replay compared 96 fitted quantities (excluding standard errors), maximum change $7.12\text{e-}6$; this is a third, distinct count.

A development calibration reused 40 datasets (ten per parent distribution) across six methods. Its revised numerical regression passed 240 dataset/method units. This does not reproduce the original simulation-performance study; 348 small-sample descriptive summaries do not support coverage or heavy-tail performance claims. Calibration failures, fixes and costs remain in the state ledger and original receipts, outside the paper's empirical claim.

Final candidate acceptance

The subsequent frozen upstream-retrieval candidate completed its first uninterrupted acceptance in 30.135660 elapsed seconds: setup 5.868547, numerical chain 21.825473, diagnostic data 0.913200 and plots 1.515001 seconds. It verified the two pinned archive hashes, reconstructed all annual inputs, and preserved every original flag in 120 scalar comparisons including standard errors (113 pass, seven fail). All 48 printed endpoints and 96 solver checks passed. Maximum scalar change from the saved primary values was $1.510259887\text{e-}5$. [Full receipt](#), [frozen candidate](#), [setup receipt](#), [uninterrupted chain](#).

The run used existing Python 3.12, NumPy 2.3.5, SciPy 1.18.1 and Matplotlib 3.10.7 installations. No fresh dependency installation or another-machine test was performed. Historical R-reference calculations were not rerun. This acceptance validates the exact local candidate; successful retrieval is not a data-licensing determination or a guarantee of publication.

Primary empirical table comparisons

Each cell is computed value (published target). An asterisk retains a failed original scalar check; it is not removed after the separate reference-optimizer explanation. Values shown to six decimals; original full-precision receipts determine pass/fail.

Table S2. Plymouth

Quantity	All adjusted	All naive	Reduced adjusted	Reduced naive
Location μ	128.776231 (128.77) *	126.519186 (126.52)	129.552359 (129.55)	128.586175 (128.59)
Scale σ	18.805577 (18.81)	25.503721 (25.5)	17.657293 (17.65) *	17.777388 (17.78)
Shape ξ	0.000972 (0)	-0.282264 (-0.28)	0.041516 (0.04)	0.035579 (0.04)
SE(μ)	4.401715 (4.4)	5.526858 (5.53)	4.605521 (4.6) *	4.544832 (4.55) *
SE(σ)	2.633604 (2.63)	3.997957 (4)	3.405781 (3.41)	3.607907 (3.61)
SE(ξ)	0.158454 (0.16)	0.151853 (0.15)	0.264913 (0.26)	0.268539 (0.27)
Return level 25	189.020069 (189)	180.241961 (180)	189.951266 (190)	188.809531 (189)
Lower endpoint 25	168.357944 (168)	168.514784 (169)	168.235965 (168)	167.702085 (168)
Upper endpoint 25	259.262263 (259)	214.784526 (215)	329.775308 (330)	324.265589 (324)
Return level 50	202.293699 (202)	186.838471 (187)	204.344410 (204)	202.998216 (203)
Lower endpoint 50	175.331429 (175)	174.355850 (174)	175.073327 (175)	174.643199 (175)
Upper endpoint 50	308.717954 (309)	234.772289 (235)	461.393298 (461)	453.779887 (454)
Return level 100	215.478294 (216) *	192.210969 (192)	219.053046 (219)	217.437716 (217)
Lower endpoint 100	180.843200 (181)	178.701822 (179)	180.215259 (180)	179.856700 (180)
Upper endpoint 100	370.052831 (370)	255.437053 (255)	668.474832 (668)	658.271183 (658)

Table S3. Brest

Quantity	All adjusted	All naive	Reduced adjusted	Reduced naive
Location μ	52.888175 (52.89)	52.271597 (52.27)	52.840947 (52.84)	52.568800 (52.57)
Scale σ	11.843322 (11.84)	12.088768 (12.09)	11.924809 (11.93) *	12.004302 (12.01) *
Shape ξ	-0.023815 (-0.02)	-0.030113 (-0.03)	-0.029933 (-0.03)	-0.031987 (-0.03)
SE(μ)	1.065369 (1.07)	1.072816 (1.07)	1.075964 (1.08)	1.076497 (1.08)
SE(σ)	0.736111 (0.74)	0.761623 (0.76)	0.751830 (0.75)	0.760046 (0.76)
SE(ξ)	0.044488 (0.04)	0.043859 (0.04)	0.043008 (0.04)	0.042689 (0.04)
Return level 25	89.362599 (89.4)	89.134167 (89.1)	89.213879 (89.2)	89.066065 (89.1)
Lower endpoint 25	84.021545 (84)	83.868420 (83.9)	83.947499 (83.9)	83.835431 (83.8)
Upper endpoint 25	97.593802 (97.6)	97.136512 (97.1)	97.204340 (97.2)	96.952092 (97)
Return level 50	97.017928 (97)	96.775428 (96.8)	96.756332 (96.8)	96.603635 (96.6)
Lower endpoint 50	90.342160 (90.3)	90.234267 (90.2)	90.232763 (90.2)	90.137087 (90.1)
Upper endpoint 50	108.527516 (108.5)	107.935468 (107.9)	107.839344 (107.8)	107.527902 (107.5)
Return level 100	104.490959 (104.5)	104.201875 (104.2)	104.087674 (104.1)	103.919689 (103.9)
Lower endpoint 100	96.254977 (96.3)	96.174517 (96.2)	96.104718 (96.1)	96.020110 (96)
Upper endpoint 100	120.055319 (120.1)	119.269431 (119.3)	118.984015 (119)	118.590944 (118.6)

Reader's reproducibility guide

Use the [upstream-retrieval candidate](#). It reconstructs the two empirical applications from version-pinned source archives; it does not reproduce the original simulation study. [Manuscript](#), [numerical supplement](#), [Tables S2-S3](#).

Setup and run

Use Python 3.12 with NumPy 2.3.5, SciPy 1.18.1 and Matplotlib 3.10.7. requirements-plot.txt includes the numerical dependencies. For a new installation, create a virtual environment and install that file; the acceptance check used existing installations rather than testing a fresh installation. From a clean copy containing the files listed in candidate-manifest.json, run:

```
python setup_sources.py
```

```
python run_all.py --output-dir results
```

```
python prepare_diagnostics.py --results-dir results --output results/diagnostics.json
```

```
python render_diagnostics.py --input results/diagnostics.json --output-dir results/figures
```

Setup downloads evmissing 1.0.2 and Renext 3.1-4 from the exact URLs in [setup_sources.py](#), checks their pinned SHA256 hashes, and reads named archive members without executing R. The Brest event CSVs come from Renext; its annual reference is decoded from evmissing's BrestSurgeMaxima. No prebundled annual JSON is needed. The annual aggregators independently compare 27 Plymouth and 162 Brest rows to the source objects.

A completed setup creates sources/ and data/. Existing setup/output directories are rejected to avoid overwriting evidence. On an error, inspect setup-receipt.json, results/receipt.json and the stage logs. The numerical chain has a 120-second overall cap; each application fit is capped at 200,000 calls/30 CPU seconds/60 elapsed seconds, and each audit at 5,000 calls/15 CPU seconds/30 elapsed seconds. Partial output is not a successful run. Plots use English labels and Matplotlib's bundled DejaVu Sans; no macOS-only font is required.

Verified result and boundaries

The first exact-candidate acceptance passed in 30.14 seconds, including upstream retrieval and plots. [Acceptance receipt](#) records all 120 scalar comparisons including standard errors: 113 passed, seven original discrepancies remained, and no original flag changed. All 48 printed endpoints and 96 independent solver checks passed. The frozen code was unchanged throughout. See the [source manifest](#), [chain receipt](#) and [render receipt](#).

Earlier stopped and separately continued runs remain in the private research record as historical evidence; they are not part of this public package. They are not rewritten as uninterrupted successes. The final acceptance does not re-execute historical R-reference calculations or the simulation study, and does not prove fresh-install or second-machine portability. Data are retrieved from the pinned original package archives under their published terms; see [data provenance and notices](#). The archived setup receipt records the technical test's contemporaneous licence_cleared=false flag, not an editorial or legal determination. Final public release is separate from the successful technical check.